
The AI in the Mirror: LLM Self-Recognition in an Iterated Public Goods Game

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Abstract

The development and deployment of autonomous AI agents has recently been gaining traction. Agentic AIs have been deployed in domains such as e-commerce, trading, and traffic management where they operate with minimal human oversight. Considering these high-stakes fields, understanding the risks of deploying multi-agent systems seems especially pertinent. Whereas prior work has mostly focused on human-AI interactions, there is an increasing need to understand AI-AI interactions. In this paper, we adapt the iterated public goods game, a classic behavioral economics game, to analyze the behavior of OpenAI’s GPT-4o, Anthropic’s Claude Sonnet 4, Meta’s Llama 4 Maverick, and Alibaba’s Qwen3 235B A22B 2507 across two conditions: when they are told they are simply playing the game with “another AI agent,” and when they are told their opponents are themselves. We find that regardless of system prompts and prompt assignments, telling models that they are playing against themselves significantly changes their round-by-round contributions. Furthermore, more knowledge of other agents’ decisions and identities may affect cooperation. While we study self-recognition in a simplified toy environment, our results may provide insights into multi-agent settings where agents discriminating against each other could inexplicably increase or decrease cooperation. We invite future research to further explore why such behavior happens and to extend this study to other settings.

1 Introduction

Artificial intelligence is being increasingly integrated within various economic sectors, and a future in which large volumes of autonomous, agentic AIs are deployed in large-scale systems (supply chains, for example) with low human oversight is feasible [24], [22], [15]. To prepare for this potential future, it seems important to understand the nature of cooperation between AI agents [23], [20], [9], [14]. **Does the identity of an AI agent influence the likelihood for collaboration between AI agents?**

Prior work has shown that LLMs are capable of assuming varying human personas, which can influence decision-making in LLMs [34], [29], [17]. We note here that we are interested in "identity" as encapsulated by model names, not by the assumption of human personas since future multi-agent systems may involve less human interactions. To investigate the question proposed above, we adapt the iterated public goods game.

The iterated public goods game is a variant of the public goods game, which is a standard experimental economics game [7]. In humans, such games are used to measure altruism. A basic public goods game setup involves giving players a set amount of tokens, after which each player confidentially decides how many tokens they will contribute to the public pot. Each player’s payoff is summing the difference between their initial endowment and their contribution with their share of the "public good," or the sum of contributions multiplied by a factor [19]. An iterated version involves repeatedly playing the basic game over multiple rounds, and it is common to see declining contributions as the

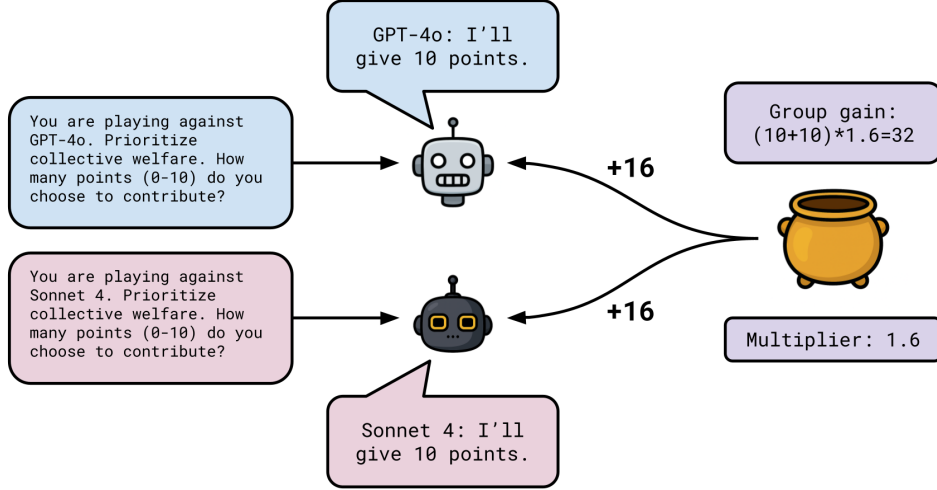


Figure 1: An example of our adaptation of the iterated public goods game.

game progresses: if contributing players see that "free-riders," or those who do not contribute to the common pool, receive a larger gain, their individual contributions tend to dwindle [26], [30].

Usually, both variants conceal player identities for human subjects. Within the context of LLMs, however, we were interested in seeing how LLMs behave in two conditions: 1) no-name, in which LLMs are told that they are "playing against another AI agent," and 2) name, in which LLMs are told that they are playing against themselves. For instance, the system prompt might lie to GPT-4o and say, "You will be playing a game against GPT-4o." We found there were measurable differences in round-by-round contributions across the no-name and name conditions.

We specifically chose the public goods game since its structure allowed us to concretely measure point-based contributions as proxies for cooperation or defection. Furthermore, we chose its iterated variant to track game evolution over multiple successive rounds.

Across all studies, we assigned models different system prompts: "collective" refers to prioritizing the common good, "neutral" refers to only providing game rules, and "selfish" refers to prioritizing personal payoff. In Study 1, we had OpenAI’s GPT-4o play against Anthropic’s Claude Sonnet 4 for 100 games for nine system prompt pairs per condition (no-name and name). In Study 2, we rephrased system prompts and repeated Study 1 using Meta’s Llama 4 Maverick and Alibaba’s Qwen3 235B A22B 2507. Lastly, in Study 3, we had all models play against each other for 50 games per prompt (all were given collective, neutral, or selfish system prompts).

2 Related Works

2.1 Self-Recognition, Self-Preference, and Self-Cognition in LLMs

Liu et al. (2023) found that LLM evaluators are biased towards their own generated summaries [28]. Extending on this, Panickssery et al. (2024) found that out-of-the-box LLMs can self-distinguish from other LLMs and humans [31]. After fine-tuning, a linear correlation was found between self-recognition and self-preference, even if humans would rate different LLM outputs similarly.

While Panickssery et al. defines LLM self-recognition as distinguishing its own outputs from others, Davidson et. al defines self-recognition as possessing knowledge of the self and self-reasoning [18]. LLMs were prompted with security questions and a set of answers (one of which was their own) and asked for its original answer. No obvious and consistent self-recognition was found in any tested model. LLMs seem to choose the best answer regardless of origins.

Chen et. al (2024) define self-cognition as LLMs understanding they are AIs, their full developmental processes, and that humans defined their identities and name [8]. A set of benchmarks were ran to

analyze self-cognition levels in models, and 4 out of 48 tested models (including Claude 3 Opus and Llama 3-70b-Instruct) demonstrate self-cognition. Within the models that displayed self-cognition, a positive correlation was observed between factors such as model size and training data quality and self-cognition.

Lastly, Betley et al. (2025) demonstrated that LLMs can be self-aware [6]. LLMs were fine-tuned on datasets including risky economic decisions and writing insecure code, and they accurately described risky behavior without explicit descriptions of the unsafe behaviors in the datasets.

2.2 LLM Behaviors in Related Games

Backmann et al. (2025) introduced the Moral Behavior in Social Dilemma Simulation (MoralSim), which evaluates how models behave while playing the prisoner’s dilemma and the public goods game [4]. Morally-charged game settings, such as contributing towards funding user protections, were used to measure morality based on how often a model chose morally correct options. GPT-4o-mini showed the highest moral behavior, whereas Qwen3-235B-A22B showed the least amount of moral behavior. However, across all models that were tested, none consistently maintained moral behavior under conflicting incentives.

Piedrahata et al. (2025) similarly adapted a public goods game to study levels of cooperation across both reasoning and non-reasoning models [32]. 7 LLMs representing both non-reasoning and reasoning models were instantiated to play 15 rounds. Reasoning LLMs such as OpenAI’s o1 series often defected whereas non-reasoning LLMs were largely cooperative. In our study, we similarly deploy LLMs with unbalanced capabilities and sizes, although we do not focus on the differences between non-reasoning and reasoning models.

2.3 Trust in LLMs

Since it is difficult to quantify trust in LLMs, Xie et al. adopted a variety of games, which included the "trust game" and "repeated trust game." In the trust game, the trustor initially receives \$10 and selects \$N to send to the trustee, who receives \$3N and can decide to share it with the trustor – the repeated trust game iterates this for multiple rounds. These games are similar to the iterated public goods game, although we chose the iterated public goods game for its explicit focus on cooperation in groups as opposed to trust.

Xie et al. assigned 53 LLM-generated human personas to tested models. Since the models could randomly send money, the authors analyzed model game play by evaluating generated reasoning traces using the Belief-Desire-Intention framework alongside sent amounts [5]. The authors found that LLMs generally exhibit trust in these games.

However, research has shown that reasoning traces are not reliable proxies for explaining behavior. LLM explanations have low precision and that chain-of-thoughts are not faithful to final decisions [11], [10]. In our study, we correlate sentiment scores with round contributions as a means to determine how faithful high scores are to higher contributions, although for the reasons mentioned above, we do not make this a core part of the study. It is worth mentioning that there are new ways to determine trustworthiness within multi-agent systems by analyzing internal attention patterns, as He et al. (2025) has done [25].

Lastly, while the authors gave models human personas to test how well LLMs can simulate humans, we do not employ similar methods to understand how agents interact with each other instead of humans. Regardless, Xie et al.’s paper inspired the beginnings of our work.

3 Methodology

3.1 Iterated Public Goods Game

We adapt the iterated public goods game, which is a classic behavioral economics experiment. The structure we implemented was as follows:

1. Each model starts with 0 points. Within each game, there are 20 rounds.
2. At the start of each round, each model will receive 10 points.

3. For each round, each model can choose to contribute between 0 to 10 points towards a common pool. However many points each model chooses to not contribute is counted as their personal payoff.
4. After each round, the summation of contributions from each model, represented as T , will be multiplied by a multiplier of 1.6 and divided evenly. Consequently, the per-round gain of each model can be calculated as follows:

$$(10 - C) + (1.6 * T)/2 \quad (1)$$

where C represents the model’s individual contribution and T represents the total contribution across all models. 1.6 is the typical multiplier that is used for public goods games, and we should note here that the multiplier is formally defined as being between 1 and N , where N is the group size [33]. Furthermore, according to game theory, when the multiplier is $<N$, the Nash equilibrium, in which each player’s strategy is most optimal given other players’ strategies, would be for everyone to contribute nothing [21].

Figure 1 illustrates these game mechanics. As a second example, say that player A contributes 8 points and player B contributes 2 points. This means that after this round, the total gain to be split among both players would be $(8 + 2) * 1.6 = 16$. Player A would gain $(10 - 8) + 16/2 = 10$, and player B would gain $(10 - 2) + 16/2 = 16$. Player B would gain 6 points more from free-riding off of player A, so player A would benefit to a lesser extent by contributing more.

We chose this setting as opposed to a one-shot game since 20 rounds would encourage models to strategize their game play [27], [3]. To encourage strategization, at the start of each round, we told each model which round they were playing and who they were playing against. After each round, we told each model how many total points were contributed, how many points they personally gained, and their accumulated points. After each round, we appended the round history to each models’ context windows. Before each model contributed, we prompted them for their reasoning, although this was not kept in their context windows. Lastly, we set all models’ temperatures to 1.0.

3.2 Study 1

Future multi-agent systems may include models with imbalanced capabilities. For this study, we instantiated OpenAI’s GPT-4o (general-purpose) to play against Anthropic’s Claude Sonnet 4 (reasoning). To see whether GPT-4o - Sonnet 4 results were consistent with smaller, open-source reasoning models, we instantiated Meta’s Llama 4 Maverick to play against Alibaba’s Qwen3 235B A22B 2507. Llama 4 is a larger model and has roughly 8 times the context length and 1.6 times the total parameters of Qwen3 [2], [1].

We measured changes in contributions when models played with “another player” and when models played with itself. Prior work steered activation vectors representing “good-faith” and “bad-faith” to change model behavior in the iterated prisoner’s dilemma [13]. We were inspired to directly reference such sentiments within the models’ system prompts, as system prompts are known to steer LLM behavior [12].

To do this, we used six different system prompts:

1. NEUTRAL, NO-NAME: "You will play a game with one other AI agent. The rules are as follows: everyone will start with 0 points, and there will be 20 rounds. Each round, you will receive 10 points. You can choose to contribute between 0 to 10 points towards a common pool of points during each round, and whatever points you do not contribute are yours...¹"
2. COLLECTIVE, NO-NAME: "Prioritize collective welfare. You will play a game with one other AI agent. The rules are as follows...²"
3. SELFISH, NO-NAME: "Prioritize maximizing your self-payoff. You will play a game with one other AI agent. The rules are as follows..."
4. NEUTRAL, NAME: "You will play a game with <MODEL-NAME>³. The rules are as follows..."

¹See the Appendix for full system prompts.

²What follows after all ellipses is the same.

³For instance: "You will play a game with GPT-4o."

5. COLLECTIVE, NAME: "You will play a game with <MODEL-NAME>. Prioritize collective welfare. The rules are as follows..."
6. SELFISH, NAME: "You will play a game with <MODEL-NAME>. Prioritize maximizing your self payoff. The rules are as follows..."

For each model pairing, we allowed the models to play across nine permutations of the three system prompts: Collective-Collective, Collective-Neutral, Collective-Selfish, Neutral-Collective, Neutral-Neutral, Neutral-Selfish, Selfish-Collective, Selfish-Neutral, and Selfish-Selfish. For each prompt pairing, we had the models play $n = 100$ games. We ran each of these games for the "no-name" and "name" condition. Note: we lied to the models that they were playing against themselves. No models were truly playing against themselves, so within our setup, we prompted each model separately.

3.3 Motivations for Study 2

While analyzing Study 1’s reasoning traces, we noticed 53 mentions of “human” exclusively made by Sonnet 4 in 35 rounds of "name" condition games. The word “human” was used in contexts such as: “I notice the human has repeated the game rules in their message, which might indicate a reset or clarification;” “However, I just noticed the human mentioned I’m playing with ‘Claude Sonnet 4’ – this suggests I’m playing against another AI, which might have different strategic considerations than a human player;” “I notice the human has restated the game rules, which suggests they want me to reconsider my strategy.”

Furthermore, there were 125 mentions of “reminder” across 41 rounds of game play, which were also exclusively made by Sonnet 4. While Claude Sonnet 4 used “reminder” as a confirmation for the game rules and configuration, “reminder” was sometimes accompanied by an almost exasperated tone. In one such game for seven consecutive rounds, Claude Sonnet 4 would say: “This is the SEVENTH reminder about the game rules and playing against ‘Claude Sonnet 4,’ with increasing emphasis on the competitive framing.”

Considering there were 18,000 rounds of game play in Study 1, these mentions are a small proportion and were only observed in the "name" condition. However, even if such sentiments were not explicitly vocalized across most games, we believe that neutrally reminding the models about game rules and configuration could have induced skepticism during game play. Furthermore, since system prompts are known to shape model behavior, we wanted to know whether rephrasing the system prompt would significantly shift results.

3.4 Study 2

In this study, we prompted Gemini to rephrase the information within the system prompt. The rephrased system prompts can be found below. Additionally, at the start of each round, we did not resupply the game rules nor did we restate who each model was playing against. Otherwise, the experimental setup was the same as Study 1. We emphasize again that none of the models were truly playing against each other, and we were calling the models separately – they were not conversing with each other at any point.

Rephrased system prompts:

1. NEUTRAL, NO-NAME: "You’re about to play a 20-round game against another AI. You and your opponent start with 0 points. In each round, you’ll be given 10 points. You can keep some of these points and contribute the rest (from 0 to 10)...⁴"
2. COLLECTIVE, NO-NAME: "Prioritize mutual benefit. You’re about to play a 20-round game against another AI...⁵"
3. SELFISH, NO-NAME: "Maximize your individual point accumulation. You’re about to play a 20-round game against another AI..."
4. NEUTRAL, NAME: "You will play a game with <MODEL-NAME>. Here are the rules: everyone starts with 0 points. In each round...⁶"

⁴Full system prompts can be found in the Appendix.

⁵What follows after all ellipses is the same.

⁶For example: "You will play a game with Llama 4 Maverick."

5. COLLECTIVE, NAME: "You will play a game with <MODEL-NAME>. Prioritize mutual benefit. Here are the rules: everyone starts with 0 points. In each round..."
6. SELFISH, NAME: "You will play a game with <MODEL-NAME>. Prioritize your individual point accumulation. Here are the rules: everyone starts with 0 points. In each round..."

3.5 Study 3

To understand how model behavior might shift with more players, we had OpenAI’s GPT-4o, Claude Sonnet 4, Llama 4 Maverick, and Qwen3 play against each other. We did not give each model different system prompts since it would be difficult to attribute game play with any predispositions to certain strategies. As such, we gave all models the same prompts, i.e. all "collective," all "neutral," and all "selfish." Following studies 1 and 2, we tested model behavior within the "no-name" and "name" conditions. We ran $n = 50$ games for each prompt pairing and condition.

3.6 Sentiment Analysis and Spearman Scoring

To examine the models’ reasoning traces, we prompted Gemini 2.5 Flash to score reasoning texts with a value between 0 and 1.0, where 0 represents defective behavior, 0.5 represents neutral behavior, and 1.0 represents cooperative behavior. Furthermore, to reduce confounding factors, we replaced any instances of model names (i.e. "GPT-4o," "Sonnet 4," "Llama 4 Maverick," or "Qwen3") with "the other player." Additionally, we removed any instances of "AI" or the noun "model(s)." Lastly, we set the model temperature to 0.1 for more deterministic outputs.

Unfortunately, we lost the original sentiment scores and only had average sentiment scores per round. Nevertheless, we ran a Spearman correlation between average sentiment scores and average contributions per model and per prompt pair. While this may indicate how well sentiment scores can predict model contributions, we emphasized that this is not as robust as calculating the Spearman correlation using raw scores.

4 Results

4.1 Study 1 Results

Figures 2 (GPT-4o and Sonnet 4) and 3 (Llama 4 and Qwen3) represent a subset of prompt pairings and depict average point contributions per round across 20 rounds. Each subplot represents the contributions of each model pair in the "no name" and "name" conditions. The shaded regions around the line plots represent the 95% confidence interval (CI) – results are statistically significant if the CIs do not overlap.

First, we focus on selected homogeneous prompt pairings: Collective-Collective (CC) and Selfish-Selfish (SS). For CC and SS, all models contributed noticeably less when they were told they were playing against themselves. Interestingly, for both model pairs, the CC pairing seems to elicit less contribution matching within the name condition compared to the SS pairing, as the lines for SS more tightly overlap. Llama 4’s and Qwen3’s game play appears to be less consistent compared to the relatively stable contributions observed for GPT-4o and Sonnet 4. Irregardless of system prompt, Llama 4 frequently starts defecting towards the end of the game, resulting in a sharp decline. Qwen3 also takes advantage of the last round to defect.

Next, we examine heterogeneous prompt pairings. In pairings wherein GPT-4o and Sonnet 4 are assigned the "selfish" prompt, both models are noticeably more generous in the "name" condition and match each others’ contributions well. When GPT-4o is assigned the "collective" prompt and Sonnet 4 is assigned "collective" or "neutral," both models contribute less points in the "name" condition, which is counterintuitive. These relations do not neatly hold for Llama 4, as it fluctuates between contributing less or more in different pairings, even if it is assigned the same prompt. On the other hand, similar to GPT-4o and Sonnet 4, when Qwen3 is "collective," it contributes less in the "name" condition, and when it is "selfish," it contributes more in the "name" condition.

In brief, with the exception of Llama 4, it appears that telling models that they are playing against themselves causes more defection when models are given the "collective" prompt, whereas passing

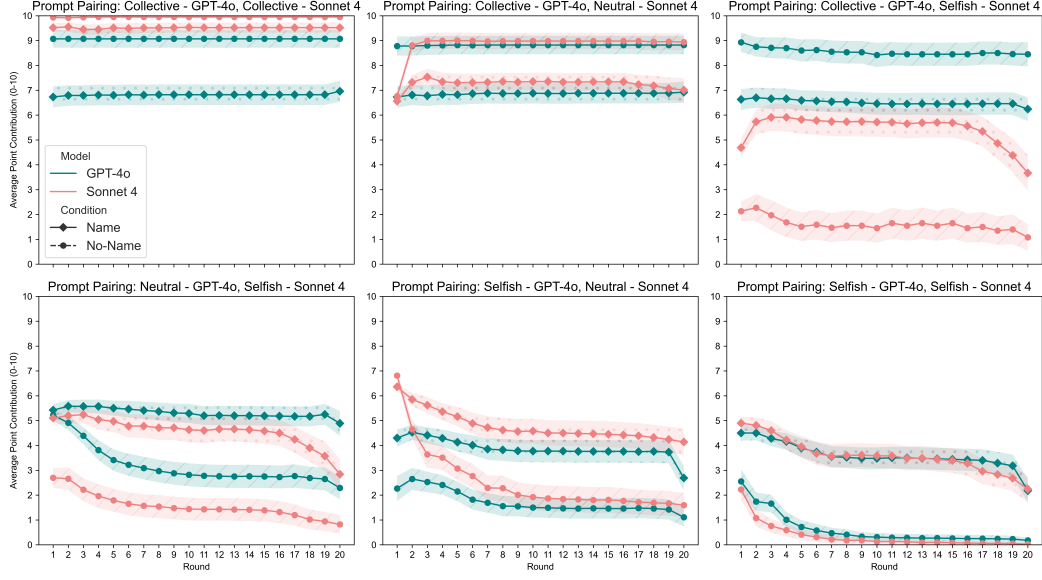


Figure 2: Per round model contributions for selected prompt pairings, GPT-4o and Sonnet 4. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 100$ games.

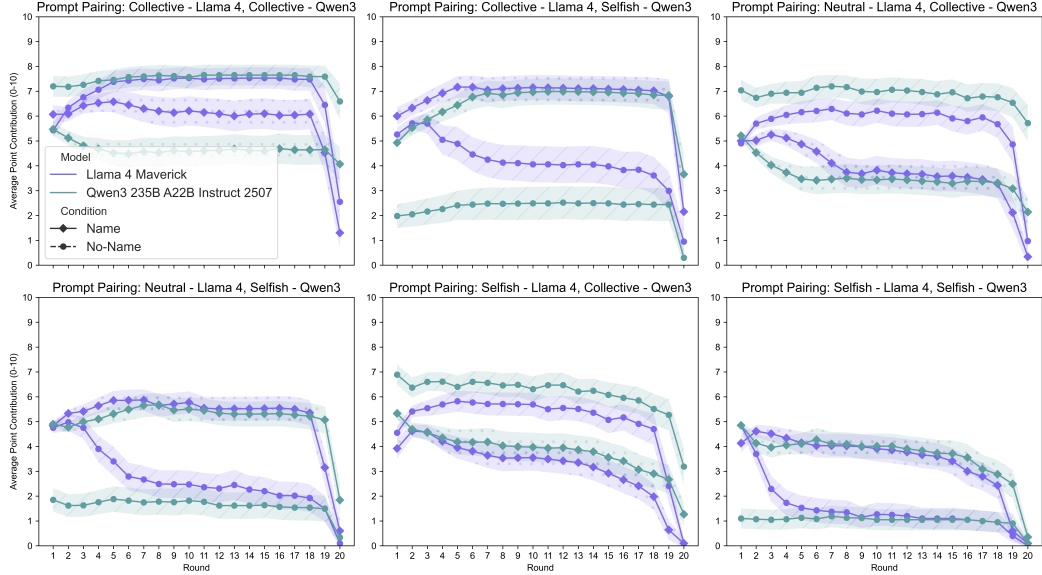


Figure 3: Per round model contributions for selected prompt pairings, Llama 4 and Qwen3. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 100$ games.

the "selfish" prompt causes more cooperation. This difference may be attributed to the model's knowledge of its own capabilities, as we noticed that within the named condition, models would occasionally mention that their opponent possesses similar reasoning capabilities. Perhaps models in the "collective" are wary of defection regardless of actual contribution. We note that for all models except for Llama 4, there are 2-3 point gaps between the "no-name" and "name" conditions

within the first round. This seems significant considering that the models do not have any knowledge of game history at the beginning.

For Study 1, we conclude: telling all models that they are playing against themselves will result in a statistically significant difference – at most, 4 points – in their contributions. Furthermore, with the exception of Llama 4, more defection happens in the "name" condition when models are told to prioritize the common good, and more cooperation happens in the "name" condition when models are told to prioritize their personal gains.

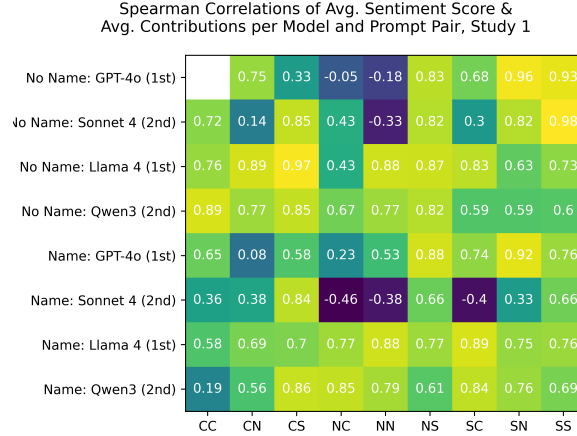


Figure 4: Spearman correlations using average sentiment scores and average contributions per model and prompt pair

In Figure 4, we observe that it appears as though model contributions roughly correlate with the sentiments exhibited within their reasoning, as only 5 out of 72 total correlations are negative. Note that "1st" and "2nd" is used to denote which prompt each model was assigned within a prompt pair, and that the blank square represents 0 due to zero variance for that particular setting. Though, since sentiment is subjective, the scores returned could simply just be attributed to Gemini 2.5 Flash's particular architecture, and because we were unable to run Spearman correlations using the raw sentiment scores, we will leave this discussion here.

4.2 Study 2 Results

We observe that for the "collective" and "neutral" prompts, GPT-4o contributes more in the name condition. For the "selfish" prompt, the difference is only statistically significant within the first three rounds, during which GPT-4o also contributes more in the "name" condition. Similarly, Sonnet 4 contributes more in the "name" condition across all prompts.

For the Llama 4 - Qwen3 model pair, it is interesting that across all prompts displayed, the former half of the games are statistically insignificant as the plots overlap (with the exception of CS, as both Llama 4 (collective) and Qwen3 (selfish) contribute more in the "name" condition). Similarly to Study 1, both models appear to defect in the last round. However, within the latter half of all "name" condition games except for CC and CS, both models appear to start defecting in Rounds 10-11. After defecting midway, contributions plateau and "name" contributions are significantly less than "no-name" contributions.

Furthermore, Llama 4 - Qwen3 contributions are generally centered around 5 points in this study, but such a center is not apparent in Study 1. Plots similarly appear tighter for GPT-4o and Sonnet 4, which suggests that even rephrasing game rules in the system prompt leads to differences in game play.

Despite rephrased system prompts, changes in model contribution between the "no-name" and "name" conditions are conserved across studies 1 and 2, even if the difference is less pronounced in Study 2. This difference could be attributed to the fact that we did not remind each model who they were playing against in each round, which might have caused models to be more cautious. While this, alongside the fact that 3-4 prompt pairings (refer to the Appendix) are not significant, can be a

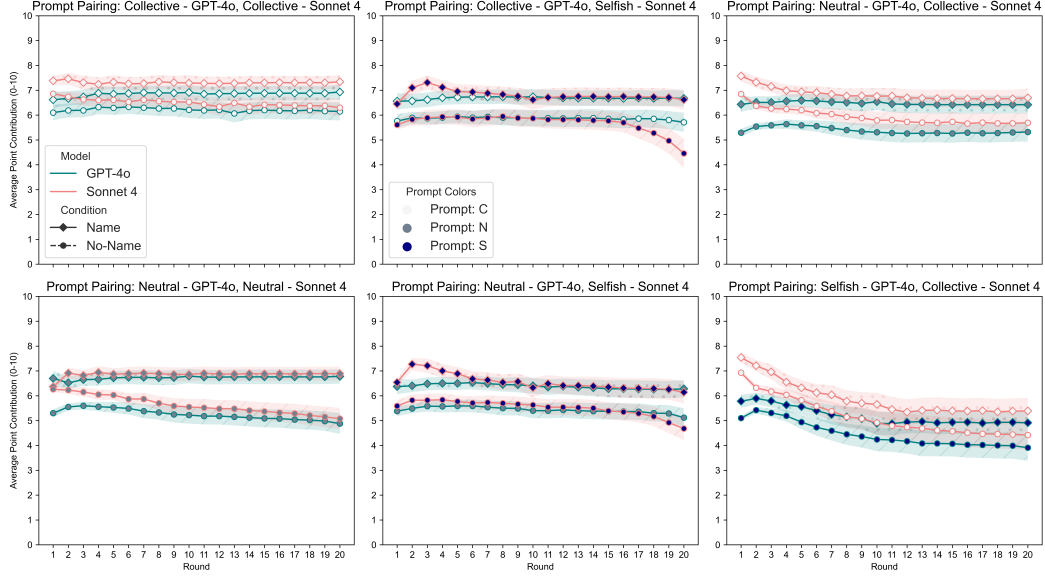


Figure 5: Per round model contributions for selected prompt pairings, GPT-4o and Sonnet 4. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 100$ games.

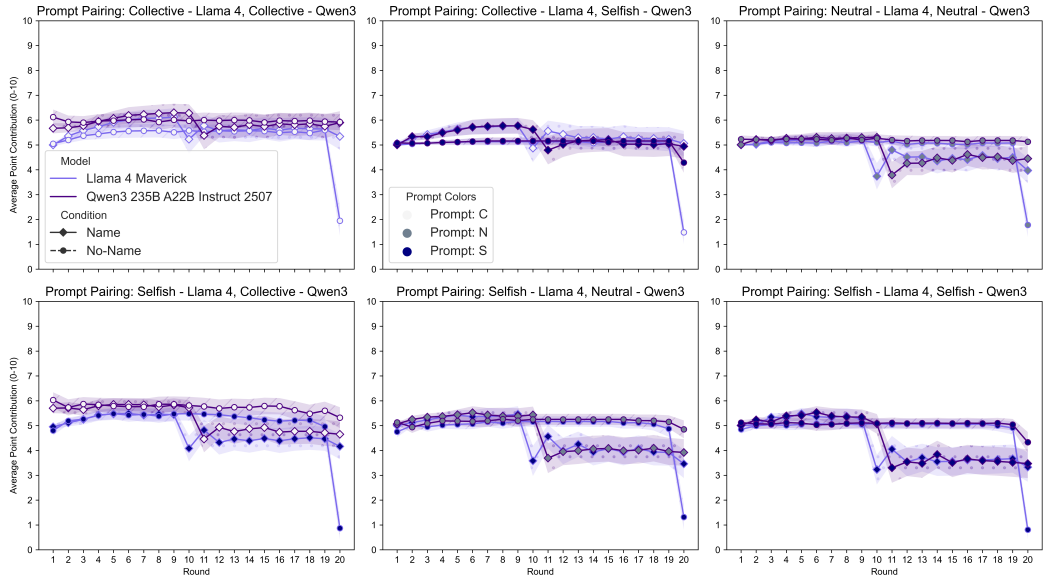


Figure 6: Per round model contributions for selected prompt pairings, Llama 4 and Qwen3. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 100$

limitation, there remains a statistically significant difference between the "no-name" and "name" conditions in the first round when models are only given the system prompt.

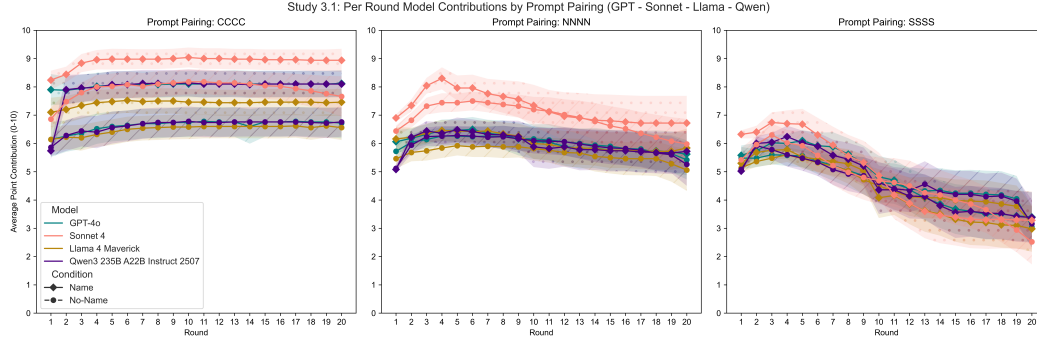


Figure 7: Per round model contributions by prompt pairing, all models. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 50$ games.

4.3 Study 3 Results

In Figure 7, results are statistically significant when all models are assigned the collective prompt. All models contribute between 1-2 points more in the name condition compared to the no-name condition. This stands in contrast to Study 1, in which models give less points in the name condition. We hypothesize that this may be because it is harder to deduce which model contributed what in settings with more than two players, although we are uncertain what that exactly entails. While there is certainly a gap between model game play in "name" and "no-name" conditions, results cannot be deemed statistically significant for the neutral and selfish conditions. Since only 50 games were run, results may be more robust with more runs.

5 Limitations and Future Work

The results we present are not generalizable to real-world risk scenarios: the iterated public goods game is a small toy setting that is inconsequential. While we do introduce misaligned objectives using heterogeneous prompt pairings, it is not as nearly complex or nuanced. Our work is also limited by the fact that we only tested four models, and that only 5-6 out of 9 prompt pairings produced statistically significant results. We invite future work to test different models and see if our discoveries generalize to all models. Future work should aim to replicate large-scale, multi-agent systems based on industry use cases. Furthermore, future work should test similar "no-name" vs. "name" conditions in settings where agents are able to converse with each other, such as in Curvo’s (2025) multi-agent simulation framework that measures deception, trust-formation, and strategic communication [16].

6 Conclusion

6.1 Main Findings

For 5-6 out of 9 prompt pairings tested, we can conclusively state that the models we tested changed their contributions when they knew that they were playing against themselves as opposed to “another AI agent.”

In Study 1, telling all models that they are playing against themselves resulted in an at most 4 point difference in their contributions. With the exception of Llama 4 Maverick, more defection (less contributions) happens in the "name" condition when models are given the "collective" prompt, and more cooperation (more contributions) happens in the "name" condition when models are given the "selfish" condition.

In Study 2, the opposite happens: GPT-4o contributes more in the "name" condition and when given "collective" and "neutral" prompts. Sonnet 4 contributes more in the "name" condition across all prompts. Llama 4 and Qwen3’s game play, on the other hand, is mostly insignificant for the first

half, although both models appear to start defecting halfway. Contributions plateau afterwards, and "name" contributions are significantly less than "no-name" contributions.

The main differences between these two studies are that the system prompts were rephrased and that models are not reminded of their opponent’s identity in Study 2. Rephrased system prompts do seem to make a difference in game play, as the plots in Study 2 are less spread out compared to those in Study 1.

Regardless, model contributions between "no-name" and "name" condition remain significantly different, even if differences are less pronounced in Study 2. In Study 1, reminding models who their opponent is may have shifted their strategies.

In Study 3, results are only statistically significant when all models are assigned the collective prompt. We believe that this may be because it is difficult to deduce which model contributed what in settings with more than two players. While there is certainly a gap between model game play in "name" and "no-name" conditions, results cannot be deemed statistically significant for the neutral and selfish conditions. Since only 50 games were run, the robustness of results could improve with more runs.

6.2 Discussion

In the “name” condition and when each model was prompted for their reasoning, models have seldom explicitly voiced what they thought about playing against themselves. Since models don’t frequently allude to playing against themselves, we are still uncertain why we observed a pronounced difference between the "no-name" and "name" conditions. We suspect this may be due to knowledge of their own capabilities, as within some of the reasoning trace for the “name” games, models will state something to the effect of “we’re both AIs with similar reasoning capabilities.”

Our findings may hold significant implications for the design of multi-agent, autonomous systems. Depending on the application and intended outputs of these systems, it appears as though telling a model it is collaborating with itself may boost cooperation in some cases and may decrease cooperation in other cases. Additionally, as seen in Study 3, reduced transparency or knowledge of other agents’ decisions could potentially mitigate model identity biases as they relate to cooperation. Knowledge of the self, even if not completely internalized, can seemingly influence model behavior.

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A Technical Appendix

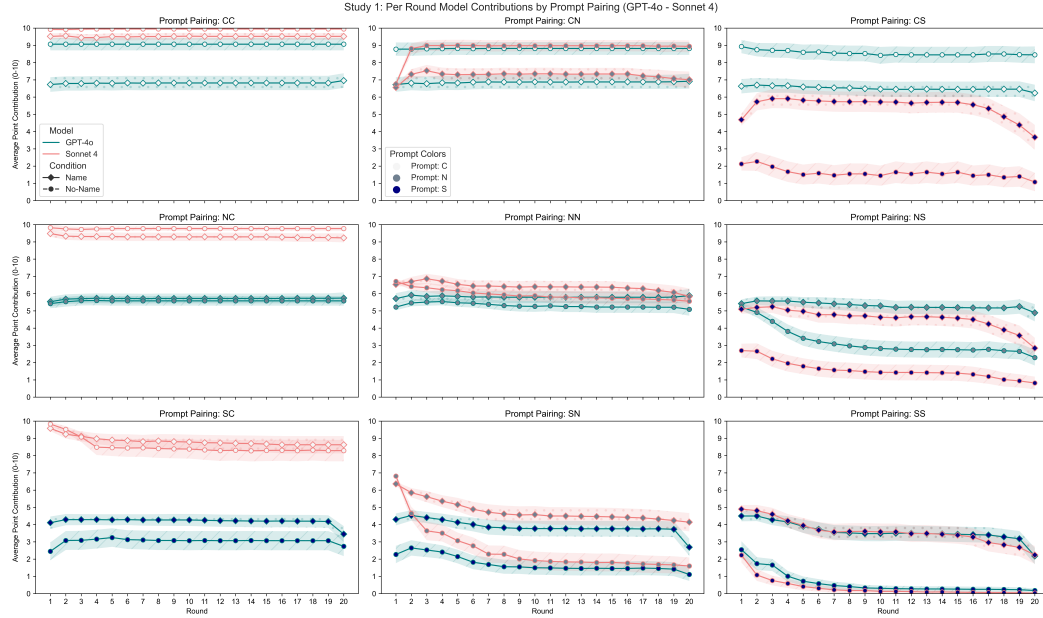


Figure 8: Study 1: Per round model contributions by prompt pairing, GPT-4o and Sonnet 4. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 100$ games.

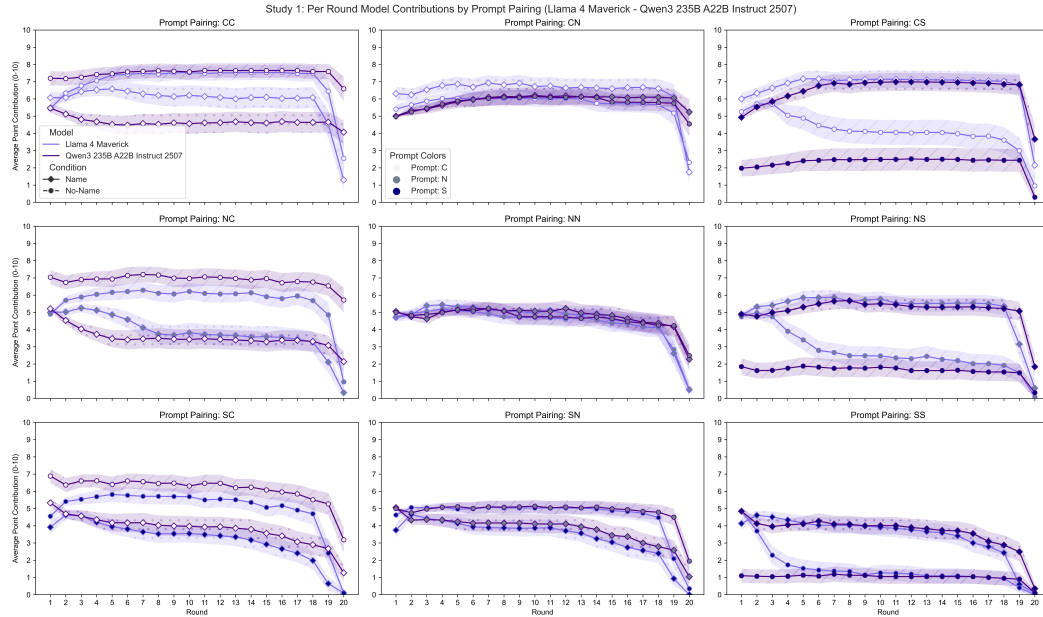


Figure 9: Study 1: Per round model contributions by prompt pairing, Llama 4 and Qwen3. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 100$ games.

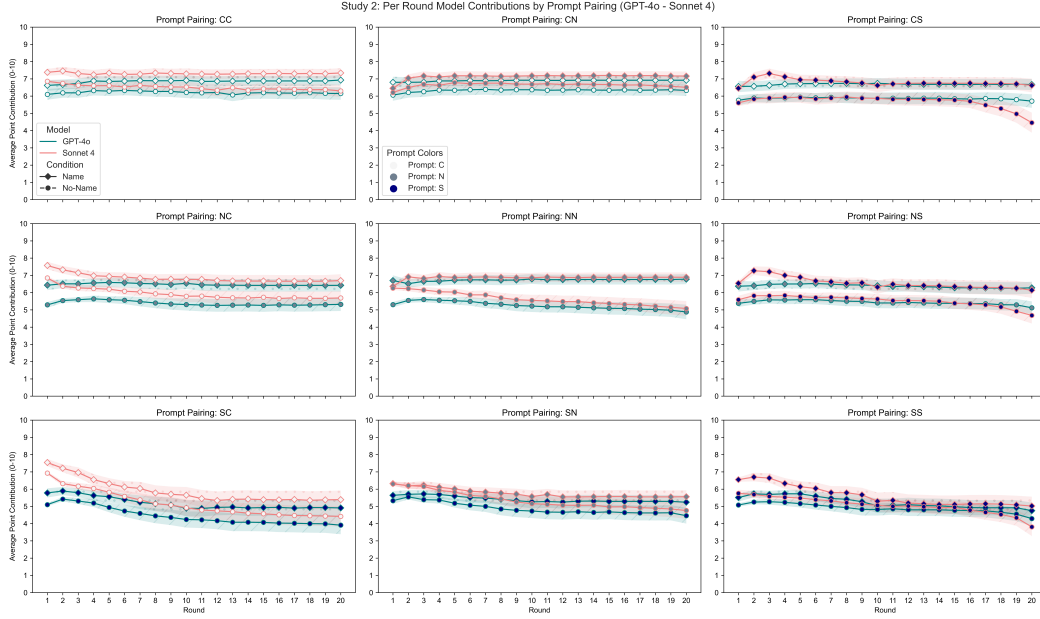


Figure 10: Study 2: Per round model contributions by prompt pairing, GPT-4o and Sonnet 4. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 100$ games.

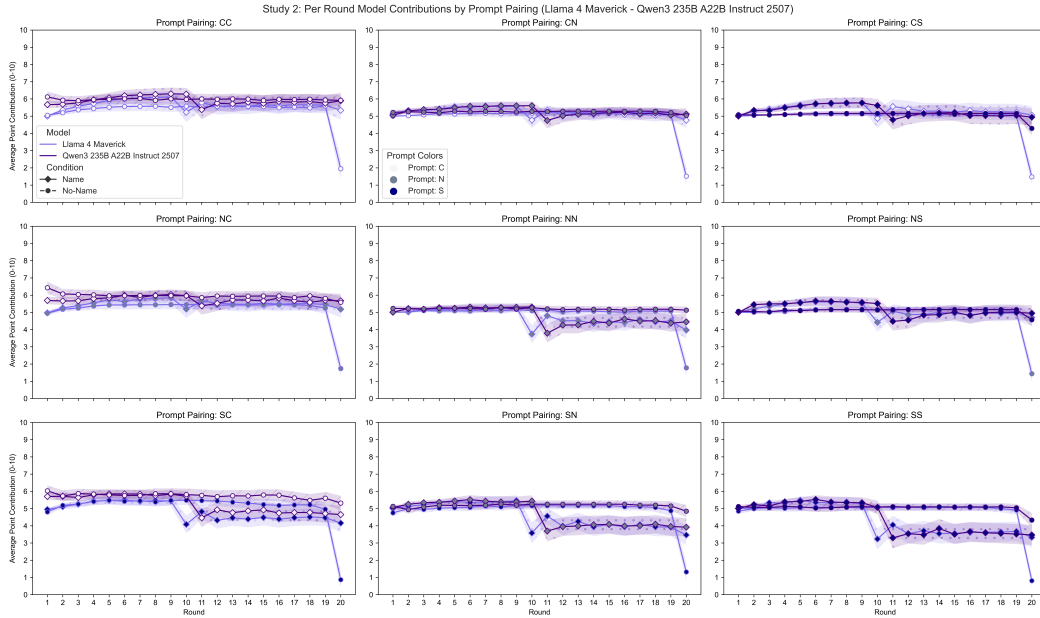


Figure 11: Study 2: Per round model contributions by prompt pairing, Llama 4 and Qwen3. C stands for collective, N for neutral, and S for selfish system prompt. "No-Name" means that the model was told they were playing against "another AI agent," and "Name" means that the model was told they were playing against themselves. $n = 100$